

On the $\mathbb{Z}_2$ invariant in the Kane-Mele model

A review introducing K-theory

Student: César Lévano

Universidad Nacional de Ingeniería
Facultad de Ciencias
Escuela Profesional de Física

Noviembre 29, 2025

- ① Introduction
- ② The Kane-Mele model
- ③ Vector bundles and K-theory
- ④ Brillouin torus
- ⑤ Time reversal symmetry
- ⑥ Conclusions and perspectives
- ⑦ Bibliography

1. Introduction

Introduction

$\mathbb{Z}_2$ Topological Order and the Quantum Spin Hall Effect

C.L. Kane and E.J. Mele

Dept. of Physics and Astronomy, University of Pennsylvania, Philadelphia, PA 19104

The quantum spin Hall (QSH) phase is a time reversal invariant electronic state with a bulk electronic band gap that supports the transport of charge and spin in gapless edge states. We show that this phase is associated with a novel $\mathbb{Z}_2$ topological invariant, which distinguishes it from an ordinary insulator. The $\mathbb{Z}_2$ classification, which is defined for time reversal invariant Hamiltonians, is analogous to the Chern number classification of the quantum Hall effect. We establish the $\mathbb{Z}_2$ order of the QSH phase in the two band model of graphene and propose a generalization of the formalism applicable to multi band and interacting systems.

PACS numbers: 73.43.-f, 72.25.Hg, 73.61.Wp, 85.75.-d

Quantum Spin Hall Effect in Graphene

C.L. Kane and E.J. Mele

Dept. of Physics and Astronomy, University of Pennsylvania, Philadelphia, PA 19104

We study the effects of spin orbit interactions on the low energy electronic structure of a single plane of graphene. We find that in an experimentally accessible low temperature regime the symmetry allowed spin orbit potential converts graphene from an ideal two dimensional semimetallic state to a quantum spin Hall insulator. This novel electronic state of matter is gapped in the bulk and supports the transport of spin and charge in gapless edge states that propagate at the sample boundaries. The edge states are non-chiral, but they are insensitive to disorder because their directionality is correlated with spin. The spin and charge conductances in these edge states are calculated and the effects of temperature, chemical potential, Rashba coupling, disorder and symmetry breaking fields are discussed.

PACS numbers: 73.43.-f, 72.25.Hg, 73.61.Wp, 85.75.-d

Figure 1: The analytical index of Kane-Mele [2][3]

Topological Field Theory of Time-Reversal Invariant Insulators

Xiao-Liang Qi, Taylor Hughes and Shou-Cheng Zhang

Department of Physics, Stanford University, Stanford, CA 94305

We show that the fundamental time reversal invariant (TRI) insulator exists in 4+1 dimensions, where the effective field theory is described by the 4+1 dimensional Chern-Simons theory and the topological properties of the electronic structure is classified by the second Chern number. These topological properties are the natural generalizations of the time reversal breaking (TRB) quantum Hall insulator in 2+1 dimensions. The TRI quantum spin Hall insulator in 2+1 dimensions and the topological insulator in 3+1 dimension can be obtained as descendants from the fundamental TRI insulator in 4+1 dimensions through a dimensional reduction procedure. The effective topological field theory, and the $\mathbb{Z}_2$ topological classification for the TRI insulators in 2+1 and 3+1 dimensions are naturally obtained from this procedure. All physically measurable topological response functions of the TRI insulators are completely described by the effective topological field theory. Our effective topological field theory predicts a number of novel and measurable phenomena, the most striking of which is the topological magneto-electric effect, where an electric field generates a magnetic field in the same direction, with an universal constant of proportionality quantized in odd multiples of the fine structure constant $\alpha = e^2/hc$. Finally, we present a general classification of all topological insulators in various dimensions, and describe them in terms of a unified topological Chern-Simons field theory in phase space.

Figure 2: The $\mathbb{Z}_2$ topological index [7]

2. The Kane-Mele model

Derivation of the Rashba term

Let the wavefunction:

$$\psi(\vec{r}) = \sum_i c_i W(\vec{R}_i - \vec{r}). \quad (1)$$

Given:

$$\nabla W(\vec{R}_i) = \left(\frac{W(\vec{R}_i + \vec{d}) - W(\vec{R}_i)}{d} \right) \hat{d} \quad (2)$$

the x-component of the momentum operator is:

$$\begin{aligned} \hat{p}_x \psi &= -i \partial_x \sum_i c_i W(\vec{R}_i) = -i \sum_i c_i \hat{e}_x \cdot \nabla W(\vec{R}_i) \\ &= -i \sum_i c_i \left(\frac{W(\vec{R}_i + \vec{d}) - W(\vec{R}_i)}{d} \right) \frac{d_x}{d}. \end{aligned} \quad (3)$$

Then the second-quantised momentum operator $\hat{p}_x$ is:

$$\begin{aligned} \int \psi^\dagger(\vec{r}) \hat{p}_x \psi(\vec{r}) d^3r &= -i \sum_{i,j} c_i^\dagger c_j \int \frac{d_x}{d^2} W^*(\vec{R}_i) \left(W(\vec{R}_j + \vec{d}) - W(\vec{R}_j) \right) d^3r \\ &= -i \sum_{i,j} c_i^\dagger c_j \frac{d_x}{d^2} \left[\int W^*(\vec{R}_i) W(\vec{R}_j + \vec{d}) d^3r \right. \\ &\quad \left. - \int W^*(\vec{R}_i) W(\vec{R}_j) d^3r \right] \end{aligned}$$

Considering only first neighbour terms:

$$\int \psi^\dagger(\vec{r}) \hat{p}_x \psi(\vec{r}) d^3r = -i \sum_{\langle ij \rangle} c_i^\dagger c_j \frac{d_x}{d^2}. \quad (4)$$

In a similar way we get the second-quantised operator $\hat{p}_y$:

$$\int \psi^\dagger(\vec{r}) \hat{p}_y \psi(\vec{r}) d^3r = -i \sum_{\langle ij \rangle} c_i^\dagger c_j \frac{d_y}{d^2}. \quad (5)$$

Thus the Rashba term is:

$$\begin{aligned} \gamma \int \psi^\dagger(\vec{r}) (\vec{\sigma} \times \vec{p})_z \psi(\vec{r}) d^3r &= \gamma \int \psi^\dagger(\vec{r}) (\sigma_x p_y - \sigma_y p_x) \psi(\vec{r}) d^3r \\ &= -\frac{i\gamma}{d^2} \sum_{\langle ij \rangle} c_i^\dagger (\sigma_x d_y - \sigma_y d_x) c_j \\ &= i\lambda \sum_{\langle ij \rangle} c_i^\dagger (\vec{\sigma} \times \vec{d})_z c_j. \end{aligned} \quad (6)$$

The Kane-Mele hamiltonian

Thus the hamiltonian is:

$$H_{KM} = t \sum_{\langle ij \rangle} c_i^\dagger c_j + i\lambda_{SO} \sum_{\langle\langle ij \rangle\rangle} \nu_{ij} c_i^\dagger s^z c_j + i\lambda_R \sum_{\langle ij \rangle} c_i^\dagger (\vec{s} \times \hat{d}_{ij})_z c_j + \lambda_\nu \sum_i \xi_i c_i^\dagger c_i$$

where:

- $t \sum_{\langle ij \rangle} c_i^\dagger c_j$ is the nearest neighbour hopping term
- $i\lambda_{SO} \sum_{\langle\langle ij \rangle\rangle} \nu_{ij} c_i^\dagger s^z c_j$ is the mirror symmetric spin-orbit interaction
- $i\lambda_R \sum_{\langle ij \rangle} c_i^\dagger (\vec{s} \times \hat{d}_{ij})_z c_j$ is the Rashba term
- $\lambda_\nu \sum_i \xi_i c_i^\dagger c_i$ is the Semenoff term

Dirac hamiltonian in the Kane-Mele model

We get after a Fourier transformation:

$$\mathcal{H}(\vec{k}) = \sum_{a=1}^5 d_a(\vec{k}) \Gamma^a + \sum_{a < b=1}^5 d_{ab}(\vec{k}) \Gamma^{ab} \quad (7)$$

such that

$$\Gamma^{ab} = \frac{1}{2i} [\Gamma^a, \Gamma^b], \quad (8)$$

$$\Gamma^{(1,2,3,4,5)} = (\sigma^x \otimes I, \sigma^z \otimes I, \sigma^y \otimes s^x, \sigma^y \otimes s^y, \sigma^y \otimes s^z) \quad (9)$$

and

$$d_a(\vec{k}) = d_a(-\vec{k}) \quad d_{ab}(\vec{k}) = -d_{ab}(-\vec{k}). \quad (10)$$

Invariance of the Dirac hamiltonian

The time-reversal operator is given by:

$$\Theta = i(I \otimes s^y)K. \quad (11)$$

Then

$$\Theta \Gamma^a \Theta^{-1} = \Gamma^a \quad \Theta \Gamma^{ab} \Theta^{-1} = -\Gamma^{ab}. \quad (12)$$

Thus following the condition for the time-reversal invariance of the Dirac hamiltonian:

$$\Theta \mathcal{H}(\vec{k}) \Theta^{-1} = \mathcal{H}(-\vec{k}), \quad (13)$$

we get that H_{KM} has time-reversal symmetry.

Pfaffian in the even subspace

Let $\vec{G}$ be a time-reversal invariant point, then:

$$\mathcal{H}(\vec{G}) = d_a(\vec{G})\Gamma^a. \quad (14)$$

Let $E_i(\vec{G})$ be the eigenenergy corresponding to the point $\vec{G}$ in the BZ at the band i , i.e.

$$\mathcal{H}(\vec{G}) \left| u_i(\vec{G}) \right\rangle = E_i(\vec{G}) \left| u_i(\vec{G}) \right\rangle. \quad (15)$$

Then, by Kramer's degeneracy, we have two pairs of orthogonal states with the same energy:

$$(\Theta \left| u_1(\vec{G}) \right\rangle, \left| u_1(\vec{G}) \right\rangle) \quad \wedge \quad (\Theta \left| u_2(\vec{G}) \right\rangle, \left| u_2(\vec{G}) \right\rangle). \quad (16)$$

Thus:

$$\Theta \left| u_1(\vec{G}) \right\rangle \propto \left| u_2(\vec{G}) \right\rangle \quad \wedge \quad \Theta \left| u_2(\vec{G}) \right\rangle \propto \left| u_1(\vec{G}) \right\rangle \quad (17)$$

Finally, $\Theta \left| u_i(\vec{G}) \right\rangle$ and $\left| u_i(\vec{G}) \right\rangle$ are equivalent up to a $U(2)$ rotation.

Pfaffian in the odd subspace

The $\mathbb{Z}_2$ invariant

3. Vector bundles and K-theory

Vector bundles

Definition (Family of vector spaces)

Let X be a topological space. A family of vector spaces over X is given by (V, π, X) where the mapping $\pi: V \rightarrow X$ is a continuous surjective map called *projection* such that for each $x \in X$ the inverse image $\pi^{-1}(x)$ is a finite-dimensional complex(or real) vector space.

Definition (Vector bundle)

A vector bundle over X is a family of vector spaces (V, π, X) such that for each $x \in X$ there is an open neighborhood U_x such that the restriction $V|_{U_x}$ is trivial.

Examples

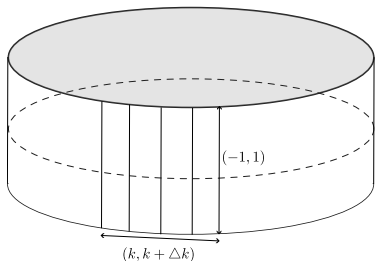


Figure 3: Cylinder

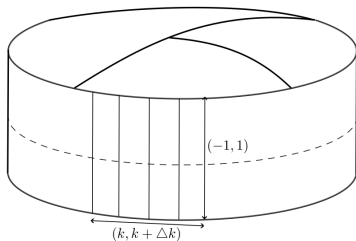


Figure 4: Möbius strip

The Grothendieck group and the K group

Let Vec_X be the equivalence class of (complex) vector bundles over the base manifold X . It has the structure of an abelian monoid.

Definition (Abelian monoid)

An abelian monoid $(\mathcal{A}, +)$ consists of:

- A commutative law of composition $+$
- An element $0 \in \mathcal{A}$ such that $0 + a = a$ for all $a \in \mathcal{A}$.

Definition (Grothendieck group)

Given an abelian monoid $(\mathcal{A}, +)$. The Grothendieck group $\mathcal{G}(\mathcal{A})$ is an abelian group that consists on the equivalence classes $\mathcal{A} \times \mathcal{A}$ given by:

$$(a_1, a_2) \sim (b_1, b_2) \leftrightarrow a_1 + b_2 + c = a_2 + b_1 + c \quad (18)$$

and the operation

$$[(a_1, a_2)] + [(b_1, b_2)] = [(a_1 + b_1), (a_2, b_2)] \quad (19)$$

4. Brillouin torus

The Brillouin torus of a honeycomb lattice

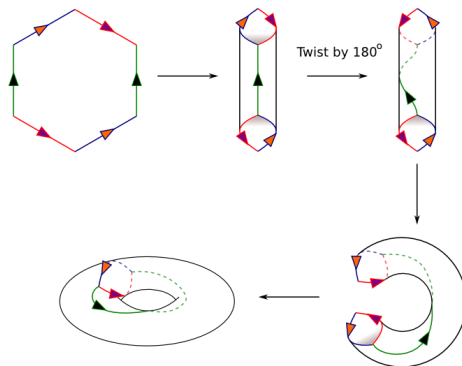


Figure 5: Gluing process of the hexagon. Extracted from Cassidy [8]

The Bloch bundle

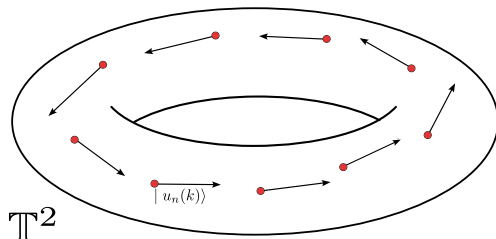


Figure 6: Bloch bundle

5. Time reversal symmetry

Brillouin torus

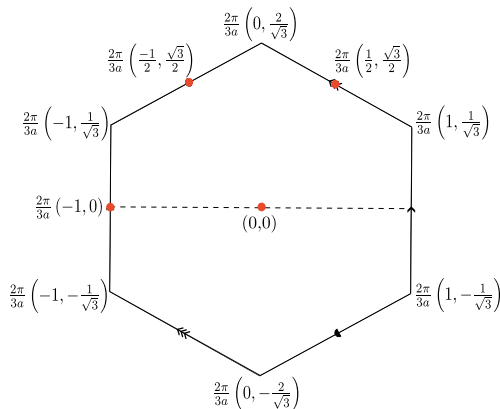


Figure 7: Time reversal invariant points of graphene

Definition (Quaternionic vector bundle)

A quaternionic vector bundle over (X, τ) is a complex vector bundle $\pi: \mathcal{E} \rightarrow X$ endowed with a homeomorphism $\Theta: \mathcal{E} \rightarrow \mathcal{E}$ such that:

- the projection π is equivariant in the sense that $\pi \circ \Theta = \tau \circ \pi$;
- Θ is antilinear on each fiber, i.e. $\Theta(\lambda p) = \bar{\lambda} \Theta(p)$ for all $\lambda \in \mathbb{C}$ and $p \in \mathcal{E}$;
- Θ^2 acts fiberwise as multiplication by -1 , namely $\Theta^2|_{\mathcal{E}_x} = -\mathbb{1}_{\mathcal{E}_x}$.

6. Conclusions and perspectives

Conclusions and perspectives



7. Bibliography

- [1] Daniel S. Freed and Gregory W. Moore.
Twisted equivariant matter.
Annales Henri Poincaré, 14(8):1927–2023, March 2013.
- [2] C. L. Kane and E. J. Mele.
 $\mathbb{Z}_2$ topological order and the quantum spin hall effect.
Physical Review Letters, 95(14), September 2005.
- [3] C. L. Kane and E. J. Mele.
Quantum spin hall effect in graphene.
Physical Review Letters, 95(22), November 2005.
- [4] Alexei Kitaev, Vladimir Lebedev, and Mikhail Feigel'man.
Periodic table for topological insulators and superconductors.
In *AIP Conference Proceedings*. AIP, 2009.
- [5] Dan Li.
Bott–kitaev periodic table and index theory, 2017.

- [6] Efton Park.
Complex topological K-Theory.
Cambridge Studies in Advanced Mathematics. Cambridge University
Press, 1 edition, 2008.
- [7] Xiao-Liang Qi, Taylor L. Hughes, and Shou-Cheng Zhang.
Topological field theory of time-reversal invariant insulators.
Physical Review B, 78(19), November 2008.
- [8] Cassidy Victor.
Flat life.
Cornell University Mathematics Department, 2009.

Gracias